

Product-Similarity Graph Clustering and Bounded Priority Heuristics for Flood-Rescue Reports: A Synthetic Held-Out Study

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Abstract. Flood-response streams may contain repeated, incomplete, and conflicting reports that must be consolidated before teams are assigned. We study a product similarity that gates temporal and contextual compatibility by geographic proximity, followed by graph clustering and a bounded, duplicate-aware priority heuristic. The mathematical result is conditional: in its finite threshold region, every retained product edge has an explicit distance bound, while a component bound additionally depends on its observed hop diameter. Evaluation uses a frozen synthetic generator with 20 development, 20 calibration, and 40 untouched test datasets. On the label-aware track, product Louvain has mean ARI 0.9237; the paired additive-minus-product difference is -0.0316 with 95% CI [-0.0393, -0.0248]. This is not universal dominance: product rejects no noise and averages 34.850 false destinations, whereas ST-DBSCAN trades lower ARI for much stronger noise rejection. Exact duplicates leave the revised priority unchanged, but a coordinated high-confidence campaign makes its drift worse than the legacy score. Independent dispatch outcomes show no benefit over the legacy policy and favor nearest-first on harm and deadlines in the tested scenarios. Consequently, this is a synthetic methodological study, not field validation or a deployment claim.

Keywords: Flood rescue · Product similarity · Graph clustering · Synthetic evaluation · Priority heuristic.

1 Introduction

Disaster-response streams combine spatial observations, evolving conditions, and repeated descriptions of the same physical incident. Treating each report as a destination can waste travel capacity; clustering solely by coordinates can merge distinct incidents that overlap geographically or split one incident reported across several sites. A coordinator also needs to inspect uncertainty and policy trade-offs before using any cluster ranking.

We investigate a pipeline that converts each report into observable attributes, constructs a spatial-temporal-contextual graph, detects communities, and computes a bounded cluster-priority heuristic. Its central composition is a product similarity: geography multiplies, and can veto, the temporal-contextual term.

Multiplicative similarities are established technology—bilateral filtering uses the same structural idea [15]—so the product form is not claimed as novel. Nor do we claim a Mercer or positive-semidefinite kernel on the Haversine domain. The narrower question is whether a fully specified product-similarity pipeline has a useful edge-localization property and how it behaves under calibrated, held-out synthetic comparison.

The contributions are:

- a complete threshold-domain statement for product and additive similarities, including a strict product edge bound and a component corollary conditional on observed hop diameter;
- a leakage-controlled protocol with disjoint development, calibration, and test seeds, two calibration tracks, tuned direct baselines, operational burden metrics, factorial diagnostics, and one sealed held-out run; and
- a duplicate-aware priority stress test and algebraically independent dispatch simulation that retain adverse results. These experiments establish exact-duplicate invariance but do not validate a rescue policy.

2 Related Work

Crisis sensing and deployment context. CrisisMMD [1] and FloodNet [12] support multimodal disaster analysis. Lightweight models such as MobileNetV3 [9], DistilBERT [14], and EmergencyNet [11] motivate on-device attribute extraction. Our generator supplies flood and urgency observations; we do not evaluate a vision/language model, device latency, energy, or field network. Edge extraction is therefore deployment context rather than an empirical contribution.

Graph and spatio-temporal clustering. Louvain [4] and Leiden [16] optimize weighted modularity without a preset cluster count. DBSCAN [7], HDBSCAN [5], and ST-DBSCAN [3] represent density-based alternatives; the latter explicitly joins spatial, temporal, and non-spatial criteria. Spatial constraints also appear in ClustGeo [6], contiguity-constrained regionalization [2], and broader spatially constrained clustering [18]. These methods make clear that geographic compactness, spatio-temporal density, and graph partitioning are not new in isolation.

Priority and humanitarian allocation. Multi-criteria allocation treats objectives and weights as policy choices [13,17]. Humanitarian logistics exposes real tension among equity, efficiency, and efficacy [8,10]. A bounded score is therefore not a validated policy: evaluation must use outcomes that do not algebraically repeat the score, and it must retain trade-offs against simpler dispatch rules.

Table 1. Positioning of the evaluated components.

Line of work	Product	Spatial control	Cluster ranking	Bound
Bilateral filtering [15]	yes	local	no	no
ST-DBSCAN [3]	no	direct	no	no
ClustGeo/regionalization [6,2]	no	explicit	no	no
This study	yes	thresholded edge	heuristic	conditional

3 Method

3.1 Observable Report Representation

A report is represented as

$$v_i = (L_i, T_i, F_i, E_i, N_i, V_i, C_i),$$

where L_i, T_i are location and time; $F_i, E_i \in [0, 1]$ are reported flood and urgency evidence; N_i, V_i are reported demand and vulnerability evidence; and $C_i \in [0, 1]$ is an interpretable provenance/corroboration heuristic. N_i and V_i are not treated as counts of unique people because the reports have no validated person-identity linkage. Latent incident identifiers, unique population, vulnerability truth, deadlines, and harm parameters are available only to the synthetic evaluator and are prohibited from clustering, tuning, and priority inference.

Confidence is computed only from observable provenance signals such as image availability and corroborating payloads. It gates E, F, N , and V consistently, but is neither a truth probability nor a misinformation detector. The recorded compact application JSON occupies 190–232 bytes (median 205.00) over 373 reports. These values exclude transport framing, HTTP/MQTT headers, TLS, authentication, retransmission, and link-layer overhead.

3.2 Product Similarity and Localization Domain

Let d_{ij} be Haversine distance and Δt_{ij} absolute time difference. Define

$$G(d) = \exp[-d^2/(2\sigma^2)], \quad T(\Delta t) = \exp[-\Delta t/\tau_t], \quad (1)$$

$$C(i, j) = \exp[-|F_i - F_j|/\tau_F - |E_i - E_j|/\tau_E], \quad (2)$$

$$w_{ij}^\times = G(d_{ij})[\beta T(\Delta t_{ij}) + \gamma C(i, j)], \quad (3)$$

where $\alpha, \beta, \gamma \geq 0$, $\sigma > 0$, and $B = \beta + \gamma$. An edge is retained exactly when $w_{ij} > \theta$; a later k -nearest sparsifier may delete but never add edges.

Theorem 1 (Product edge localization). *If $B > 0$ and $0 < \theta < B$, every retained product edge satisfies*

$$d_{ij} < r_\theta^\times := \sigma \sqrt{2 \log(B/\theta)}.$$

If $\theta \geq B$, the strict retained set is empty; for $\theta \leq 0$ there is generally no non-trivial threshold-induced finite cutoff.

Proof. Because $T, C \in [0, 1]$, $w_{ij}^\times \leq B \exp[-d_{ij}^2/(2\sigma^2)]$. In the finite region,

$$\theta < w_{ij}^\times \leq B \exp[-d_{ij}^2/(2\sigma^2)]$$

and strict monotonicity of the logarithm gives the stated radius. If $\theta \geq B$, the maximum possible weight cannot exceed the threshold. For $\theta \leq 0$, choosing $T = C = 1$ retains arbitrarily distant finite model points. Degenerate $B = 0$ cases follow from $w^\times = 0$.

Corollary 1 (Conditional component bound). *In the finite product region, a connected vertex set with at least two vertices and unweighted hop-diameter $h < \infty$ has geographic diameter $D < h r_\theta^\times$.*

The corollary follows from the triangle inequality along a path of at most h retained edges. A singleton is handled separately with $h = D = 0$. It is not an ex-ante compact-cluster guarantee: the pipeline does not enforce a useful h , and a long retained chain may approach the bound. Moreover, calibrated graph thresholds are weight quantiles, so the numerical radius is known only after the selected threshold is realized.

For the controlled additive composition

$$w_{ij}^+ = \alpha G(d_{ij}) + \beta T(\Delta t_{ij}) + \gamma C(i, j),$$

the correct contrast is threshold-dependent. If $\alpha > 0$ and $B < \theta < B + \alpha$, every retained edge satisfies

$$d_{ij} < r_\theta^+ := \sigma \sqrt{2 \log \left(\frac{\alpha}{\theta - B} \right)}.$$

At $\theta \geq B + \alpha$ the retained set is empty, while at $\theta \leq B$ there is generally no non-trivial threshold-induced geographic cutoff because the non-geographic contribution can retain arbitrarily distant pairs. Thus addition does have a finite region; claiming otherwise would overstate Theorem 1.

3.3 Duplicate-Aware Bounded Priority Heuristic

Exact-payload reports are collapsed by a fingerprint over observable location, time, measurements, provenance, and missingness fields. Observable near duplicates are coalesced by fixed geographic, temporal, measurement, confidence, and missingness tolerances. Let g index the resulting evidence families. After clipping C, F, E to $[0, 1]$, N to $[0, N_{\text{cap}}]$, and V to $[0, V_{\text{cap}}]$, the revised estimator is

$$\hat{E} = \frac{\sum_g \max_{i \in g} (C_i E_i)}{\max\{1, \sum_g \max_{i \in g} C_i\}}, \quad \hat{F} = \max_g \max_{i \in g} (C_i F_i), \quad (4)$$

$$\hat{N} = \max_g \max_{i \in g} (C_i N_i), \quad \hat{V} = \max_g \max_{i \in g} (C_i V_i), \quad (5)$$

$$\tilde{N} = \min \left\{ 1, \frac{\log(1 + \hat{N})}{\log(1 + N_{\text{ref}})} \right\}, \quad A = 1 + (\mu - 1) \tanh(\hat{V}/s), \quad (6)$$

$$R = \omega_E \hat{E} + \omega_F \hat{F} + \omega_N \tilde{N}, \quad \mathcal{P} = AR. \quad (7)$$

The maxima for N, V deliberately avoid asserting that overlapping reported counts are disjoint. With non-negative normalized weights and $1 \leq \mu \leq 2$, $\mathcal{P} \in [0, 2]$. The locked policy uses $N_{\text{ref}} = 500$ and $V_{\text{cap}} = 50$; neither these caps nor the weights have expert validation. The legacy ablation preserves report multiplicity and raw summation, including a confidence bypass for V .

4 Experimental Design

4.1 Frozen Synthetic Data

The method-agnostic Gate-1 freeze contains 80 independently generated datasets, 30229 reports, and 1280 latent physical incidents. The disjoint split is 20 development, 20 calibration, and 40 test seeds. The held-out split contains 15112 reports, ranging from 339 to 406 per dataset.

The generator varies incident geometry, report counts and dispersion, time/context overlap, missing fields, provenance, exact and near duplicates, unlinked noise, and coordinated campaigns. Each incident has latent N_{true} , V_{true} , service demand, deadline, and a piecewise lateness-harm curve. This richer internal truth supports error analysis, but the family is still designed by the authors. It cannot establish external validity for real floods.

4.2 Calibration, Baselines, and Leakage Control

Two pre-specified tracks use the same search ceilings and operational constraints. The benchmark track may use calibration labels to maximize ARI. The operational track is label-free and uses reverse-order partition stability, retained density, connectivity, diameter, and review-burden constraints. Graph thresholds are calibration-derived weight quantiles rather than shared raw thresholds across incompatible scales. Product composition comparisons additionally require matched retained fraction and mean degree.

The registry contains product and additive Louvain, a fixed seven-point convex similarity mixture (not learned multiple-kernel clustering), ST-DBSCAN, standardized DBSCAN/HDBSCAN, spatially constrained agglomerative, and same-representation Leiden/spectral diagnostics. Calibration promoted 12 feasible method-track configurations and authentically retained 8 method-track exclusions. The exclusions are not converted into poor test scores or silently removed.

The test split remained sealed until the protocol and selections were hashed. The candidate suite then ran once without seed, method, track, resume, or output filters. It produced 480 selected method-seed rows and 320 retained exclusion rows over 40 test seeds, with no prediction or scientific-seed failure.

4.3 Endpoints and Inference

Clustering co-primary families are (i) ARI on labeled reports, (ii) incident split and merge loss, and (iii) false operational destinations and operator review bur-

Table 2. Held-out means by calibration track. Split, merge, false destinations (FD), and noise rejection (NR) are for the label-aware track. Lower is better for split, merge, and FD; higher is better for ARI and NR.

Method	ARI label	ARI label-free	Split	Merge	FD	NR
Product Louvain	0.9237	0.9008	0.0938	0.0341	34.850	0
Product Leiden	0.9239	0.9007	0.0906	0.0331	34.850	0
Additive Louvain	0.8921	0.8888	0.1688	0.0341	34.875	0
Convex similarity	0.8545	0.8545	0.2375	0.0576	35.425	0
Spatial agglomerative	0.8439	0.8439	0	0.0684	31.300	0
ST-DBSCAN	0.8789	0.8438	0.00156	0.1451	0	0.8671

den over all reports. Noise rejection/absorption and destination geographic diameter are key secondary endpoints. Noise label -1 is never a destination. The standard review policy counts each unresolved report as one queue item.

Factorial diagnostics evaluate all on/off combinations of geography, time, context, and k -nearest pruning, plus confidence, vulnerability, and aggregation factors for priority. Main effects are paired factorial contrasts, not causal estimates, and must be read with interactions and unmatched cells.

Priority stress tests include exact/near duplicates, low-confidence inflation, coordinated high-confidence campaigns, and missing provenance. Dispatch uses three resource scenarios and latent outcomes that do not algebraically reuse priority, flood, vulnerability, or reported demand. We report latent harm, deadline misses, arrival mean/max/CVaR, equity gap, workload, and deadline-reached unique population; no single endpoint defines a winning policy.

Across test seeds, summaries report mean, standard deviation, median, paired bootstrap 95% CI, paired Wilcoxon signed-rank test, effect size, and denominator. Holm adjustment is applied within declared endpoint families. All ties, adverse comparisons, failures, and unavailable pairs are retained.

5 Results

5.1 Calibrated Clustering

Table 2 reports both calibration tracks. On the label-aware track, product Louvain attains ARI 0.9237 (SD 0.0185, CI [0.9180, 0.9293]). Additive-minus-product ARI is -0.0316 (CI [-0.0393, -0.0248], Holm $p = 5.09\text{e-}11$). The label-free track gives a smaller additive-minus-product difference of -0.0119 (CI [-0.0208, -0.00348], Holm $p = 0.0245$).

These ARI results coexist with material operational failures. Product Louvain has mean split loss 0.0938, merge loss 0.0341, 34.850 false destinations, and 35.400 standard-review items (95% CI [35.100, 35.700]) over 40 seeds, with registered metric-denominator total 2026 review decision units (emitted destinations plus unclustered reports). It rejects noise at rate 0. ST-DBSCAN lowers split loss by 0.0922 (CI [0.0719, 0.1125]) and improves noise rejection by 0.8671, but

Table 3. Held-out factorial main effects. Positive means the enabled factor improves the declared endpoint; CIs are paired bootstrap intervals and p_H is Holm adjusted within the factorial family.

Factorial	Factor	Mean effect	95% CI	p_H
Clustering	Geography	0.6403	[0.6334, 0.6475]	2.73e-11
Clustering	Time	0.1482	[0.1423, 0.1542]	2.73e-11
Clustering	Context	0.1264	[0.1196, 0.1333]	2.73e-11
Clustering	k -NN	0.0500	[0.0473, 0.0527]	2.73e-11
Priority	Confidence	-0.0189	[-0.0360, -0.00401]	0.1196
Priority	Vulnerability	0.0891	[0.0714, 0.1073]	2.55e-11
Priority	Aggregator	0.1219	[0.0992, 0.1463]	2.42e-07

its direction-adjusted merge-loss improvement is -0.1110. Its lower ARI therefore represents a different error trade-off, not simple inferiority.

Product Louvain and Leiden are practically tied: Leiden-minus-Louvain ARI is 2.32e-04 (CI [-8.15e-04, 0.00145], Holm $p = 0.6858$), with 35 ties. Additive also has a smaller emitted-destination geographic diameter than product by a direction-adjusted 202.09 m (CI [95.249, 332.58]). The analytic edge property therefore did not imply empirical compactness superiority.

5.2 Factorial Diagnostics

Table 3 gives pre-specified main effects. Geography, time, context, and k -nearest pruning have positive full-factorial ARI contrasts. However, all interaction orders are reported in the locked artifact and 80 density-unmatched cells remain in the analysis. The effects cannot be read as isolated causal contributions.

The confidence main effect on latent rank correlation is -0.0189 and is not significant after Holm correction ($p_H = 0.1196$). Vulnerability and aggregation effects are positive, but their interaction is also substantial: 0.1423 (CI [0.1119, 0.1732]). This supports treating the score as an interacting synthetic diagnostic, not an expert-endorsed policy.

5.3 Priority Stress Tests and Dispatch Trade-offs

At exact-duplicate multiplicity ten, the revised estimator has normalized priority drift 0, rank drift 0, and top- k churn 0. This is exact-payload invariance, not proof of accurate incident population: its remaining relative N and V errors are 0.2665 and 0.2331.

The coordinated high-confidence campaign is a declared failure. Revised normalized priority drift is 0.1688 (CI [0.1370, 0.2011]) and false-priority lift is 0.8828. The direction-adjusted revised-versus-legacy improvement is -0.1379 (CI [-0.1678, -0.1084]), with the revised estimator worse on 40 of 40 seeds. Confidence and caps bound influence but do not detect coordinated manipulation.

Table 4 reports direction-adjusted paired effects, where positive favors revised priority. Against legacy priority in the nominal dual-depot scenario, latent-harm

Table 4. Selected dispatch effects (revised policy versus comparator). Positive direction-adjusted values favor revised priority; negative values favor the comparator.

Resources	Comparator	Endpoint	Effect	95% CI	p_H
Nominal dual depot	Legacy priority	Latent harm	9.005	[-14.729, 33.082]	0.5360
Nominal dual depot	Nearest-first	Latent harm	-204.94	[-225.61, -184.90]	3.64e-12
Nominal dual depot	Nearest-first	Deadline miss	-0.2781	[-0.3391, -0.2203]	1.50e-07
Nominal dual depot	Nearest-first	Workload CV	0.0346	[0.0136, 0.0555]	0.00610
Lean Hué	Nearest-first	Latent harm	-481.39	[-520.00, -442.26]	3.64e-12
Lean Hué	Equity aging	Equity gap	-212.26	[-267.01, -158.13]	1.45e-07

improvement is 9.005 with a CI spanning zero. Against nearest-first, both harm and deadline effects are strongly negative. Nearest-first is not declared a universally preferred policy because the full artifact also contains arrival, workload, coverage, and equity trade-offs. In particular, revised priority has a positive direction-adjusted boat-workload-CV benefit of 0.0346 (CI [0.0136, 0.0555], Holm $p = 0.00610$); nevertheless, its nominal Pareto-frontier frequency is 0.2750 versus 1.000 for nearest-first.

5.4 Runtime and Representation Scope

The development-only Exp22 audit used 1 warm-up and 5 measured isolated workers with the thread limit set to 1. At 1494 events, median dense and geographic-candidate-pruned total times are 1.054s and 0.6804s. Edge counts, matrices (tolerance 10^{-9}), and partitions agree for all three measured sizes.

The spatial path prunes candidate-pair computation but still materializes a dense $O(n^2)$ compatibility matrix. The result supports exact geographic candidate pruning on the recorded one-core environment; it does not support fully sparse memory scaling, field real-time performance, or an end-to-end network-packet claim.

6 Discussion and Threats to Validity

Supported claims. The complete threshold analysis establishes a strict edge-localization property for product similarity and a conditional hop-diameter corollary. Under two frozen calibration tracks and one untouched synthetic run, product Louvain has higher ARI than the calibrated additive composition. Exact-payload deduplication makes the revised priority invariant to exact multiplicity. These statements have analytic or locked held-out support.

Necessary qualifications. Product is not uniformly superior. Leiden ties Louvain; additive has a smaller destination diameter; ST-DBSCAN markedly improves split/noise behavior while worsening merge loss and ARI; product emits many false destinations. Factorial main effects interact and include 80 density-unmatched cells. The high-confidence campaign defeats the intended priority

safeguard. Revised priority has no demonstrated latent-harm advantage over legacy and is inferior to nearest-first for harm/deadlines in the shown scenarios. These negative results prohibit claims of misinformation robustness, dispatch benefit, or operational policy validation.

Synthetic and policy validity. All reports, incidents, duplicates, and latent outcomes are generated. Multiple seeds measure variation within this generator family, not transfer to real floods. No authorized incident-level real dataset or rescue/logistics expert panel is available. Consequently, weights, caps, near-duplicate tolerances, review rules, and dispatch resources are illustrative. The arithmetic latitude/longitude centroid is metadata rather than a geodesic optimum. External collection, annotation, privacy, and expert protocols are specified, but executing them remains future work.

Method and inference validity. ARI uses labeled reports, while split/merge, noise, destinations, and review burden expose different failures. Holm adjustment controls declared families but does not eliminate generator dependence. The convex comparator is a fixed mixture grid, not learned multiple-kernel clustering. Several registered methods had no feasible configuration and remain explicit exclusions. Community detection depends on resolution and graph sparsification; the component theorem does not control h . The locked result preserves scenario-family tables, including a multimodal-focused family, but G0 does not expose a multimodal-specific claim selector. We therefore make no quantitative multimodal claim; dedicated multimodal error analysis remains unresolved.

7 Artifact Availability and Reproducibility

The single accepted held-out run identifier is:

20260728T213934Z-85f2a6686a1b-
9797f31f-x0-exp23-heldout.

Its manifest SHA-256 begins `cbedfe1c`, and its canonical result content hash begins `dda6d634`. Gate 3 records one candidate invocation. The G0 promotion includes a selector-compatible compact JSON, the complete canonical result as a lossless gzip archive, the original 448 selectors, audited data/runtime tables, a numeric claim catalog, and generated LaTeX values. Every empirical number above resolves through `\RevisionClaim` to that catalog. The one-command `reproduce.sh` workflow verifies the companion artifact package, recomputes selectors and publication values from the sealed result without re-executing the held-out experiment, runs the tests, and rebuilds this PDF. The package does not claim a public DOI or a clean immutable release commit until those external submission steps are completed.

8 Conclusion

We presented a product-similarity graph and a bounded, duplicate-aware priority heuristic for consolidating synthetic flood-rescue reports. The product form has a finite-region edge bound, but cluster compactness remains conditional on observed hop diameter. Calibrated held-out results favor product Louvain over the additive composition on ARI in both tracks, while direct baselines reveal important fragmentation, noise, diameter, and workload trade-offs. Exact duplicates are neutralized, yet coordinated high-confidence reports remain a failure, and independent dispatch outcomes do not validate the revised priority policy. Real incident annotation, expert policy validation, and fully sparse implementation are required before any deployment claim.

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